

exposures. We recommended that the Norwegian fund take a top-down, intentional approach to factor exposures. That is, Norway should practice dynamic factor investing; the fund's benchmarks should also include dynamic factors just as they do static, long-only bond and equity factors.

3. The Factor Recipe

Just as eating right requires looking through foods to their underlying nutrients, investing right requires looking through asset class labels to underlying risk exposures. Assets are bundles of factors, and assets earn returns because of their underlying exposures to factor risks. (Chapter 6 covers factor theory in more detail.)

3.1. SELECTING FACTORS

The Professors' Report to the Norwegian Ministry of Finance lists four criteria for determining which factors investors should choose. A factor should:

1. Be justified by academic research

The factors should have an intellectual foundation, and only factors with the strongest support in academic research should be included in an investor's benchmark. Research should demonstrate compelling rational logic or behavioral stories or both in explaining why the risk premiums arise. We do not need unanimity concerning the mechanism that generates the risk premium—which, if you've met financial economists, you know is impossible. (Economists are even divided over the source of the equity risk premium, for example [see chapter 8], but I am highly confident that equities have higher returns than bonds over the long run and that equities are a risk factor.) On this criterion, value-growth, momentum, and short volatility strategies are appropriate risk factors. New research can identify new factors, or qualify earlier consensus on known factors, or even disqualify factors and can inform investment policy accordingly.

2. Have exhibited significant premiums that are expected to persist in the future

Not only should we have some understanding as to why the risk premium has existed in the past, but we should have some basis for believing that it will prevail in the future (at least in the short term). Factors are systematic by definition—they arise from risk or behavioral tendencies that will likely persist (again in the short term) even if everyone knows about the factors, and many investors are pursuing the same factor strategies.¹³

¹³This chapter does not discuss the factors that are used by many (mean-variance) risk models. Most of these factors are constructed based on decompositions of a covariance matrix of (residual) returns and are not based on economic theory generating risk premiums.